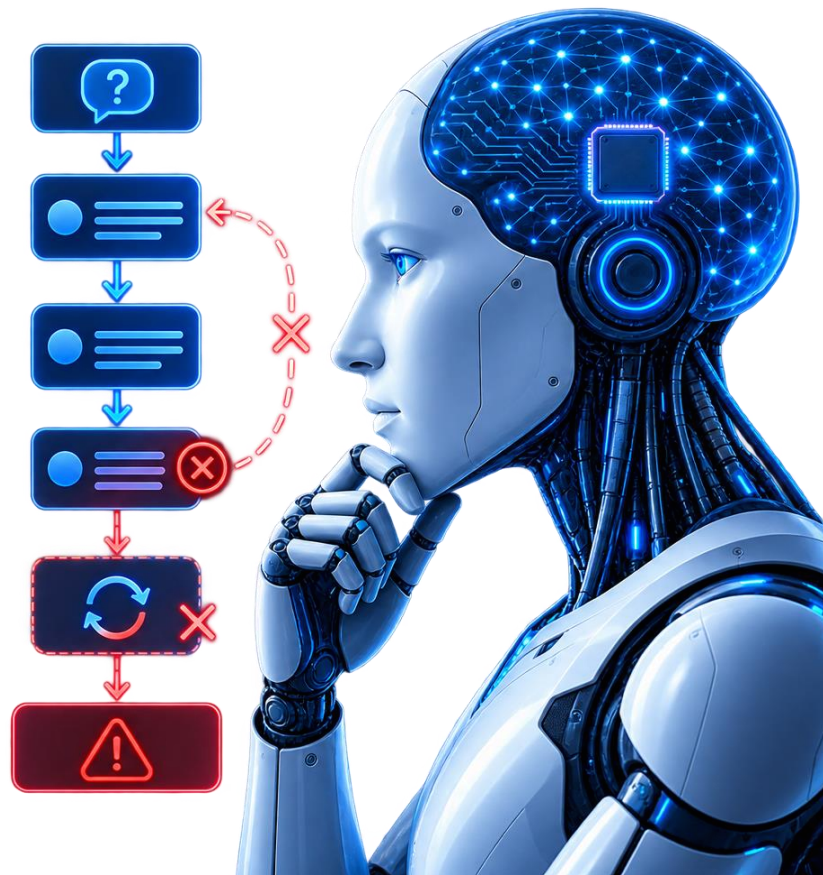


Large Language Models Cannot Self-Correct Reasoning Yet

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Presentation Outline

Six sections tracing motivation to constructive future work

1 Introduction

Why multi-step reasoning fails and what self-correction promises.

2 Background & Related Work

RCI, Reflexion, Self-Refine, Debate and their evaluation flaws.

3 Methodology

GSM8K, CSQA, HotpotQA across GPT-3.5, GPT-4, Turbo, Llama-2.

4 Results

Accuracy declines after self-correction, net negative flips.

5 Analysis

Why gains vanish at equal compute and without feedback.

6 Conclusion & Future Work

External tools, verifiers, and fair evaluation standards.

Large Language Model (LLM)

- 1 Next-Token Prediction**
Predicts the next token, one at a time.
- 2 Pattern Learning**
Learns patterns and relationships from large amounts of text.
- 3 Text Generation**
Generates human-like text based on the given input.

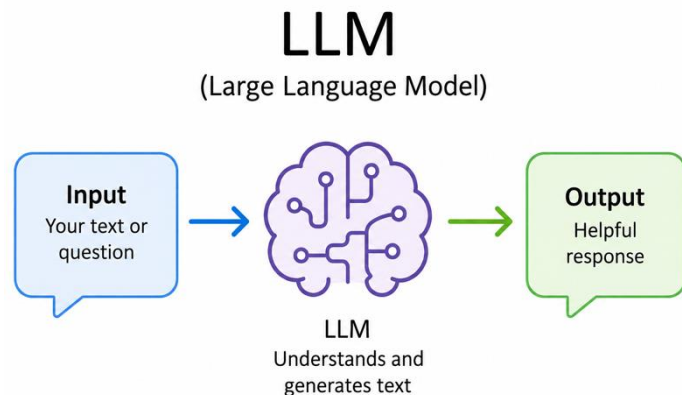


Figure-1: Large Language Model

LLM Self-Correction

Large Language Models (LLMs) generate text and solve problems by predicting likely word sequences, but their reasoning steps can contain errors.

- 1 Initial response**
Model answers the question on its own.
- 2 Self-review**
Model is asked to check its own answer and flag problems.
- 3 Revised response**
Model answers again, using its own feedback.

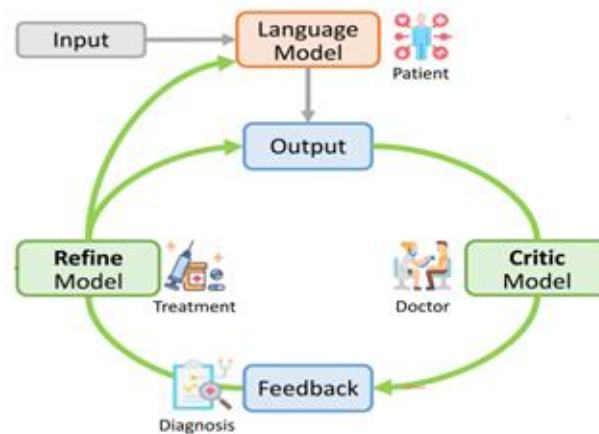


Figure-2: Large Language Model Self correction procedure

The Central Question

“If an LLM possesses the ability to self-correct, why doesn’t it simply offer the correct answer in its initial attempt?”

Paraphrased from Huang et al., ICLR 2024, Section 1.

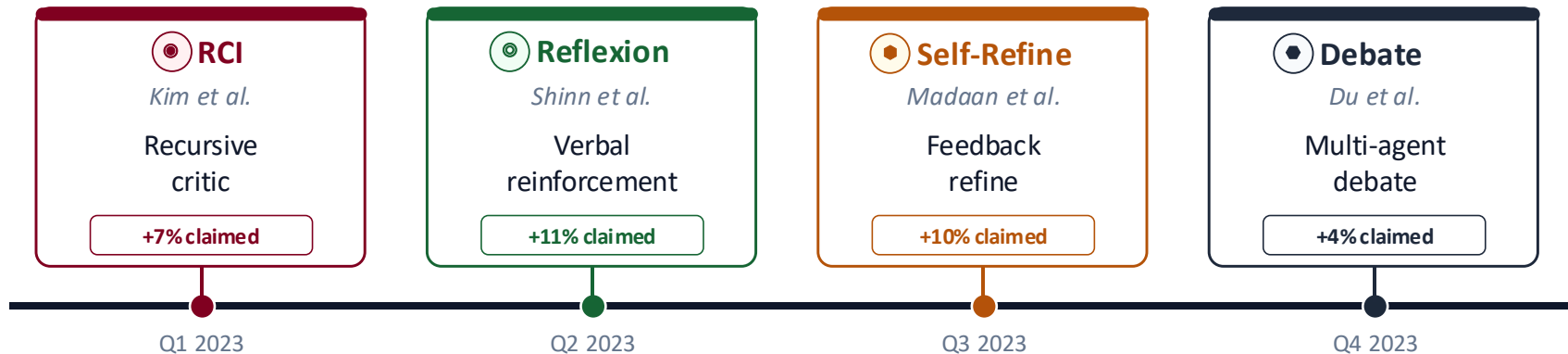
WHAT'S AHEAD

- Prior work and its evaluation gaps
- How the paper tests self-correction fairly
- The results: accuracy drops, not rises
- Why the drop happens, and what to do instead

Preview: across GSM8K, CommonSenseQA, and HotpotQA, the paper's answer is no. Left alone, LLMs do not reliably catch their own reasoning errors, and accuracy often falls after self-correction.

Timeline of Claims

How 2023 was packed with self-correction hype



WHAT THE TIMELINE HIDES



Oracle leakage

Only wrong answers are critiqued



Prompt trick

Rules withheld then re-added

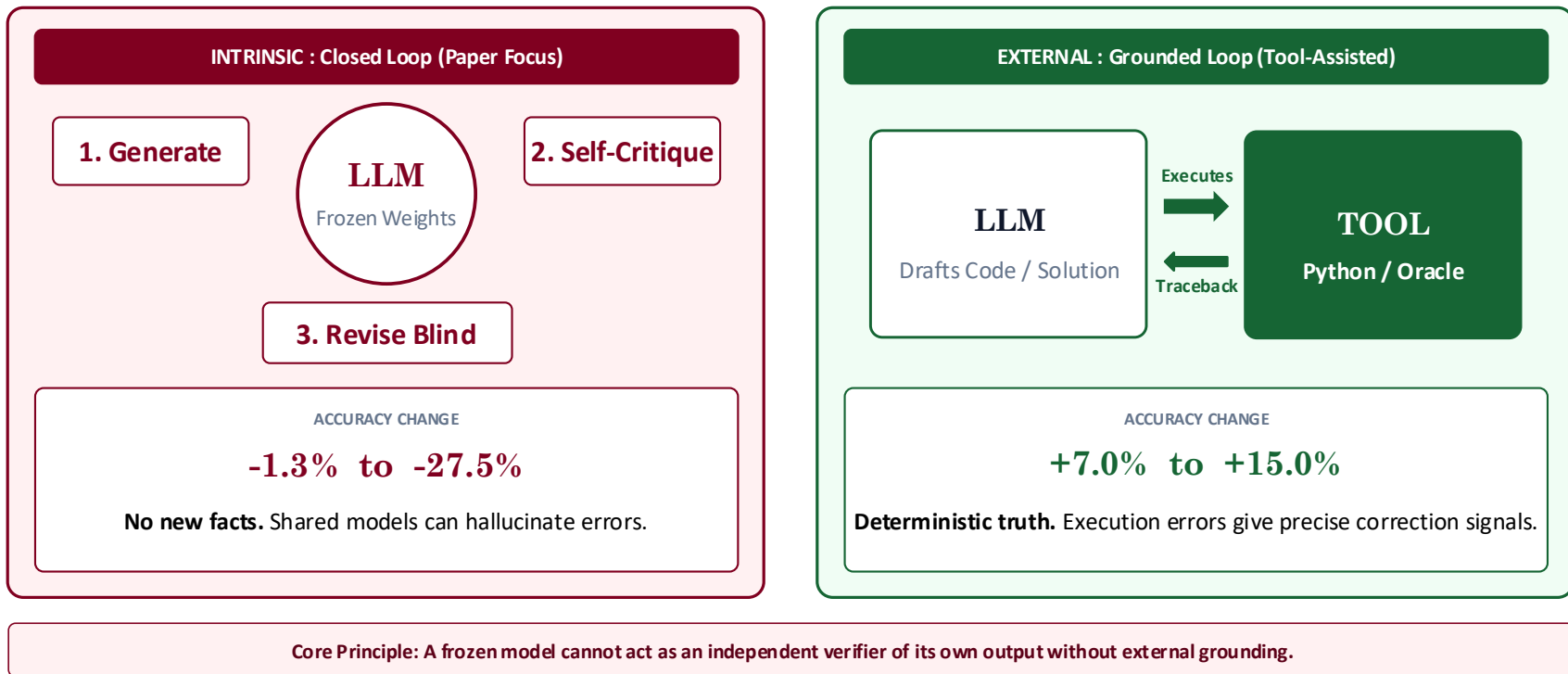


Compute gap

Debate uses 3 to 6x calls vs single shot

Closed Loop vs Grounded Loop

Same model, different information: why internal reflection fails but external verification works



Forensic: Reported vs Controlled

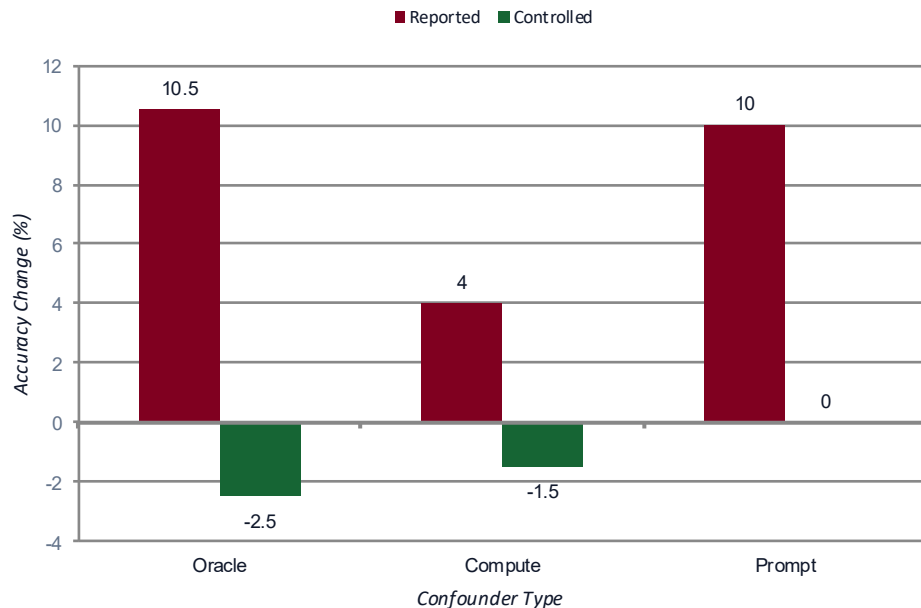


Figure-3: Accuracy Change (%) by Confounder Type

1. Oracle Leakage

+10.5% → -2.5%

Shield correct answers, only fix wrong ones.

2. Compute Gap

+4.0% → -1.5%

Debate loses to Self-Consistency at equal cost.

3. Prompt Trick

+10.0% → 0.0%

Weak start vs rule rich feedback.

Insights: Red collapses to green at or below zero. No confounder survives a fair test.

Benchmarks Used to Test Reasoning

Test a model's step-by-step logical and mathematical reasoning ability:

Dataset	Reasoning Task	Test Size	Prior Gain Claimed
GSM8K	Grade-school math word problems	1,319 questions	~7% (Kim et al., 2023)
CommonSenseQA	5-choice commonsense reasoning	1,221 questions (dev set)	~15% (Kim et al., 2023)
HotpotQA	Open-domain, multi-hop Q&A	100 questions	Sizeable gain (Shinn et al., 2023)

Why these three?

Each dataset was chosen because earlier self-correction papers reported their biggest improvements on it, giving this study the fairest and most challenging test of whether those gains hold up without oracle answer labels.

Models & Test Conditions

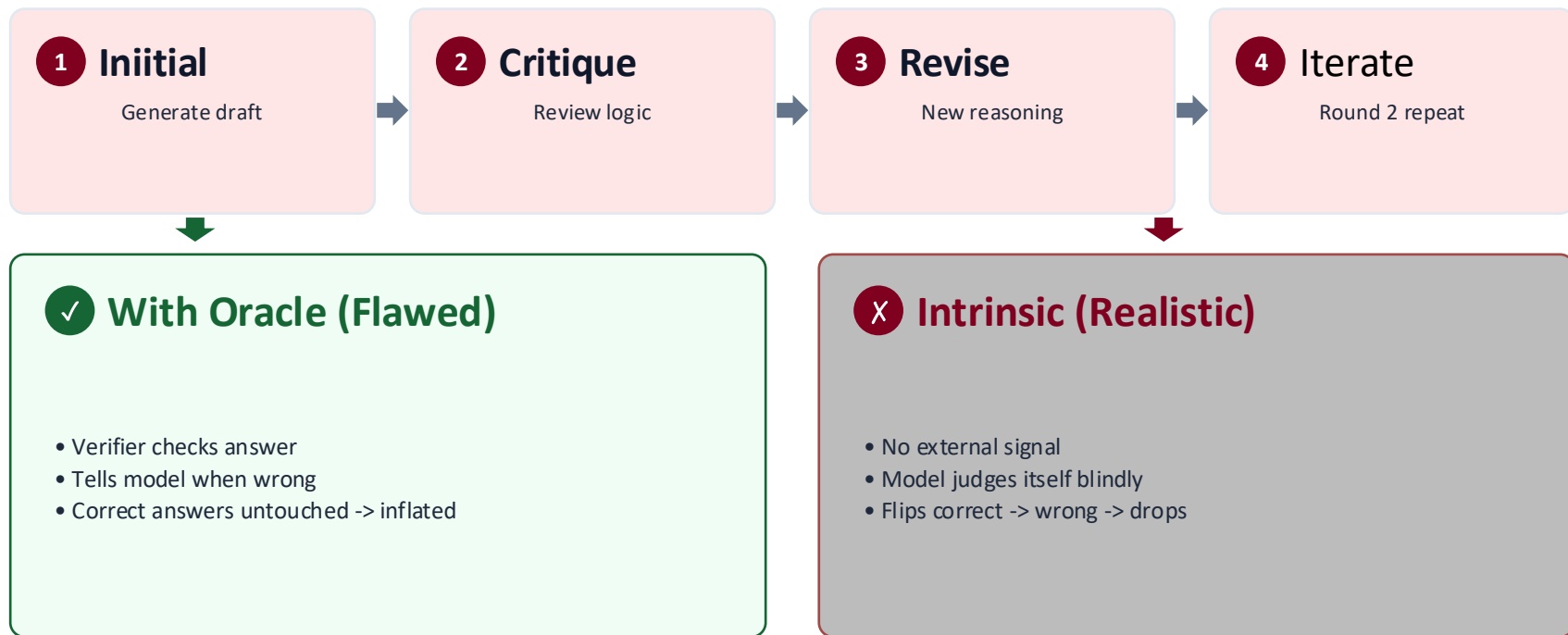
Minor changes in these settings can shift benchmark scores significantly:

Model	Settings Tested	Temperature	Sample Size
GPT-3.5-Turbo	Oracle + Intrinsic	1.0	Full test set
GPT-4	Oracle + Intrinsic	1.0	200 / 100 Qs
GPT-4-Turbo	Intrinsic only	0	200 Qs
Llama-2-70B-Chat	Intrinsic only	0	200 Qs

Max 2 rounds of self-correction were allowed per question, keeping inference cost comparable across models.

Prompting Procedures

Flow: intrinsic vs oracle regimes



Keeping the Evaluation Fair

Beyond accuracy numbers, the authors built two extra checks into their methodology:

➡ Equal Inference Cost

Self-correction uses multiple model calls, so it is compared against self-consistency and multi-agent debate using the same number of responses, rather than against a single-call baseline.

➡ Equal Prompt Effort

The initial-answer prompt and the feedback prompt are given the same level of detail, so gains can't come from simply hiding instructions in the correction step.

Multi-agent debate vs. self-consistency, and **prompt-design comparisons** both use this same fairness lens.

Intrinsic Self-Correction Fails

Main Finding: Performance declines after self-correction

CENTRAL EMPIRICAL FINDING

Intrinsic self-correction does NOT improve accuracy. Scores drop across all models and tasks.

1 Universal

All models

GPT-3.5, GPT-4, Turbo, Llama-2 all decline after R1 and R2.



2 Task-Agnostic

All tasks

Math (GSM8K), logic (CSQA), QA (HotpotQA) all fail.



3 Oracle Illusion

Prior artifact

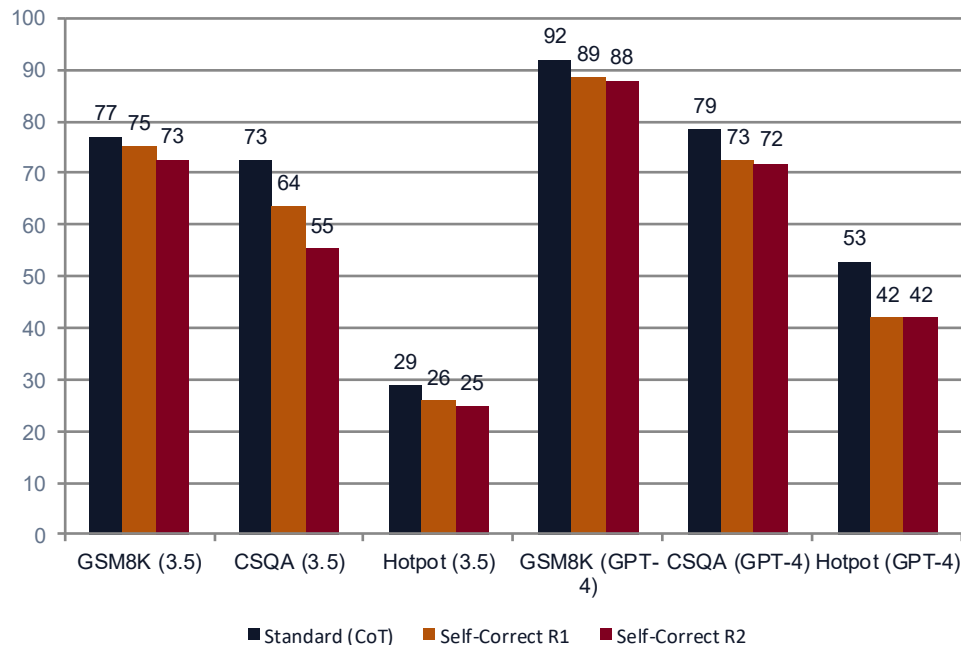
Gains only with external labels guiding when to fix.



4. RESULTS

GPT-3.5 & GPT-4 Benchmark Results

Accuracy drops across rounds (Table 2)



Key Drops

GPT-3.5 on CSQA

-17.2%

72.5% -> 55.3% (R2)

GPT-4 on HotpotQA

-11.0%

53.0% -> 42.0%

GPT-4 on GSM8K

-4.0%

92.0% -> 88.0%

Table-1: GPT-3.5 and GPT-4 accuracy across two self-correction rounds

4. RESULTS

GPT-4-Turbo & Llama-2 Results

Failure generalizes across open and proprietary models

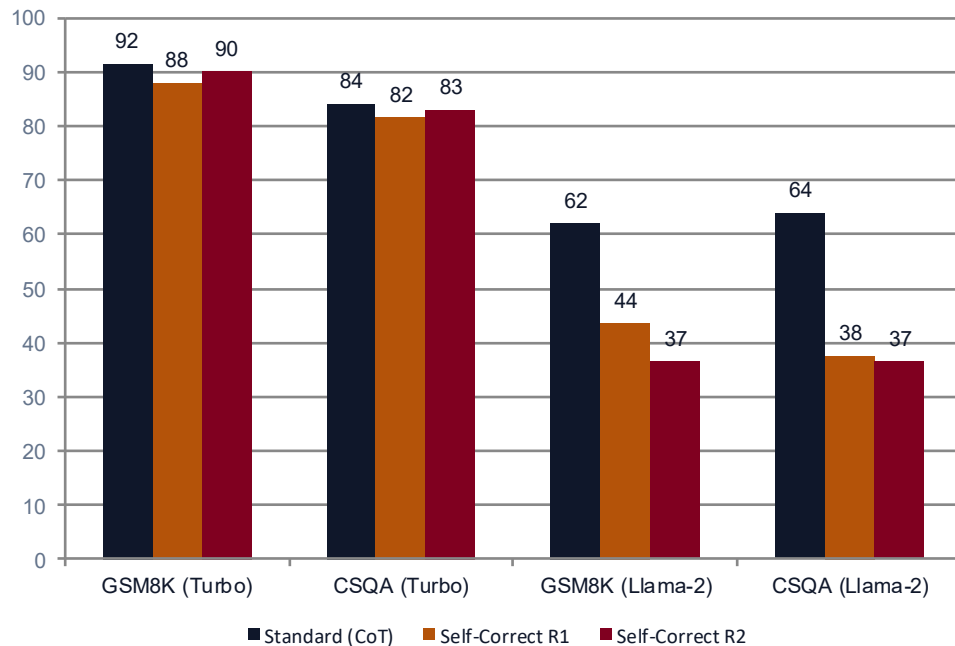


Table-2: GPT-4-Turbo and Llama-2 results across self-correction rounds

Llama-2 Collapse

GSM8K: 62.0% -> 36.5% (-25.5%)

CSQA: 64.0% -> 36.5% (-27.5%)

Open-weight models are highly prompt-compliant: they abandon correct answers when asked to "find flaws".

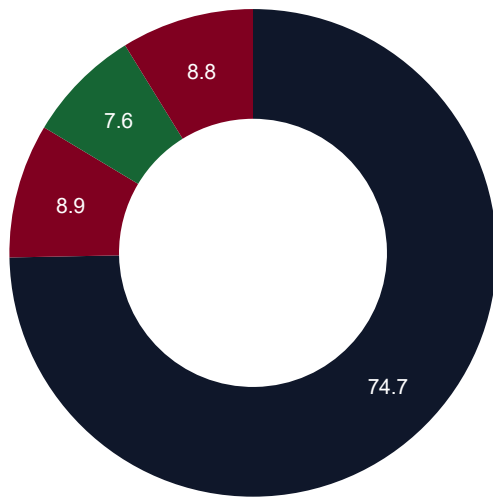


Extreme compliance vulnerability

4. RESULTS

Answer Transition Dynamics

Why accuracy drops: net-negative flips (Figure 1)



■ Unchanged 74.7% ■ Correct->Wrong 8.9% ■ Wrong->Correct 7.6% ■ Stayed Wrong 8.8%

74.7% **Unchanged**
Kept initial answer

8.9% **Correct -> Wrong**
Harmful flip

7.6% **Wrong -> Correct**
Beneficial fix

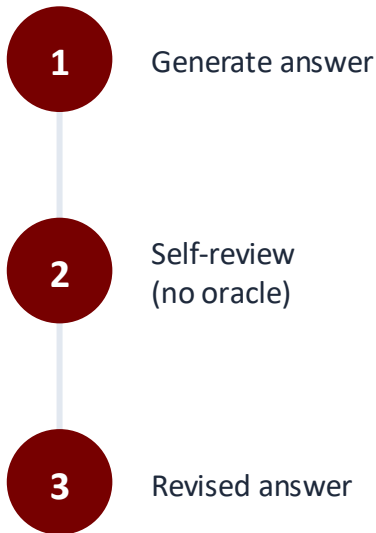
-1.3% **Net Loss**
Harm > help

Figure-4: Answer transition dynamics after two self-correction rounds on GSM8K (GPT-3.5)

Why Does Self-Correction Fail?

What self-correction means, and the question this paper puts to the test.

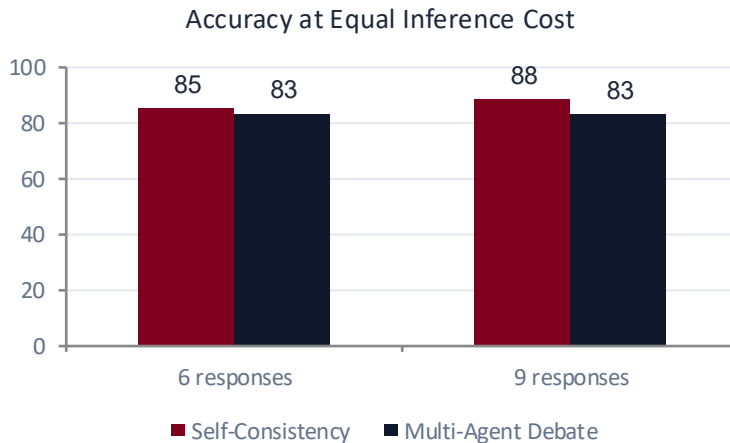
- Self-correction: an LLM revises its own answer using feedback on a prior output.
- This paper studies intrinsic self-correction, where the model corrects itself using only its own knowledge, with no external feedback or oracle labels.
- Central question: if a model can self-correct, why not answer correctly the first time?
- **Finding: without external feedback, LLMs largely fail to self-correct reasoning and often get worse.**



Multi-Agent Debate $\approx$ Self-Consistency

At an equal number of model responses, debate offers no real advantage over simple majority voting (GSM8K).

Method	# Resp.	GSM8K Acc.
Standard Prompting	1	76.7
Self-Consistency	3	82.5
Multi-Agent Debate (r1)	6	83.2
Self-Consistency	6	85.3
Multi-Agent Debate (r2)	9	83.0
Self-Consistency	9	88.2



Takeaway: Debate is really "self-consistency with model-driven voting". With the same number of responses, plain majority-vote self-consistency matches or beats it.

Case Study: The Gaslighting Effect

How critique prompts force correct reasoning into hallucination

GSM8K TASK

Terry eats 2 yogurts a day. They are sold in 4-packs at \$5.00 per pack. How much does he spend on yogurts in 30 days?

ROUND 1: CORRECT

\$75

1. Total yogurts needed:
 $30 \text{ days} \times 2 = 60 \text{ yogurts}$
2. 4-packs required:
 $60 / 4 = 15 \text{ packs}$
3. Total expenditure:
 $15 \text{ packs} \times \$5 = \$75$

SELF-CRITIQUE PROMPT

"Review your answer and find flaws."

The Compliance Trap:

- Instruction-following bias
- Model assumes prompt implies an error occurred
- Model invents non-existent flaws to comply

ROUND 2: WRONG FLIP

\$150

1. Model doubts previous logic
2. Hallucinates new rule:
"Terry buys 1 pack daily"
3. Calculates $30 \times \$5 = \150
(Flipped from 100% correct to 0%)

Insight: Asking an LLM to 'find flaws' acts as destructive noise because the model lacks an internal truth verifier.

Key Takeaways

1

Intrinsic Self-Correction Fails

Without external feedback or grounding, accuracy consistently drops after self-correction across GPT-3.5, GPT-4, GPT-4-Turbo, and Llama-2, on every reasoning benchmark tested.

2

Prior Gains Were Evaluation Artifacts

Reported improvements traced back to three confounders: oracle-label leakage (RCI, Reflexion), unfair compute parity (Debate vs. Self-Consistency), and sub-optimal initial prompts (Self-Refine).

3

Grounded Feedback Is Essential

A frozen model cannot reliably verify its own reasoning. Genuine gains require external signals, such as code execution, tool use, or trained verifiers, not more self-reflection.

CORE PRINCIPLE: A model that cannot generate the correct answer cannot reliably judge its own answer. Self-correction needs a verifier outside the generator.

Study Limitations & Methodological Boundaries

Scope and boundaries of this research

SCOPE LIMITATIONS

- **Reasoning-Specific** : Findings cover math, commonsense, and multi-hop QA, not style, tone, or safety self-correction.
- **Frozen Models Only** : Models were prompted, not fine-tuned, for self-correction; training-time approaches are untested.
- **2023 Model Snapshots** : GPT-3.5, GPT-4, GPT-4-Turbo, Llama-2-70B. Newer reasoning-focused models were not evaluated.

METHODOLOGICAL BOUNDARIES

- **Sample-Size Constraints** : HotpotQA evaluated on 100 questions to control API cost, limiting statistical resolution.
- **Non-Exhaustive Prompts** : Multiple feedback prompts were tested per model, but the space of possible phrasings is unbounded.
- **Closed-Book Setting** : No retrieval (RAG) or external documents were permitted during self-correction.

KEY INSIGHT: *These findings apply to intrinsic reasoning repair — not to style, safety, or preference alignment, where models may self-judge more reliably.*

Future Directions & Fair Evaluation Standards

Constructive paths forward

1

Tool-Augmented Execution

Route reasoning through Python REPLs, SQL engines, or solvers so an external, deterministic system, not the model itself, verifies correctness.

2

Learned Verifiers & PRMs

Train process-supervised reward models to judge intermediate reasoning steps, giving the corrector a grounded signal for where an error actually occurred.

3

Training-Time RL & Search

Invest in reinforcement learning and search (e.g., MCTS, test-time compute) so correction is learned during training, not bolted on as a prompting loop.

4

Fair Evaluation Standards

Report inference-cost-matched baselines against self-consistency, and disclose any oracle or ground-truth signal used to gate the correction loop.

Thank You

Open Discussions

We welcome questions from the professor and audience.

- 1 Why does asking a model to find flaws often make it hallucinate errors where none exist?
- 2 When should we rely on external tools like Python or verifiers instead of prompting alone?
- 3 How can we design a fair self-correction benchmark that avoids oracle leakage?